

AIIR Lab Systems for CLEF 2026 SimpleText

Ensemble and Multi-Agent Pipelines for Scientific Text Simplification

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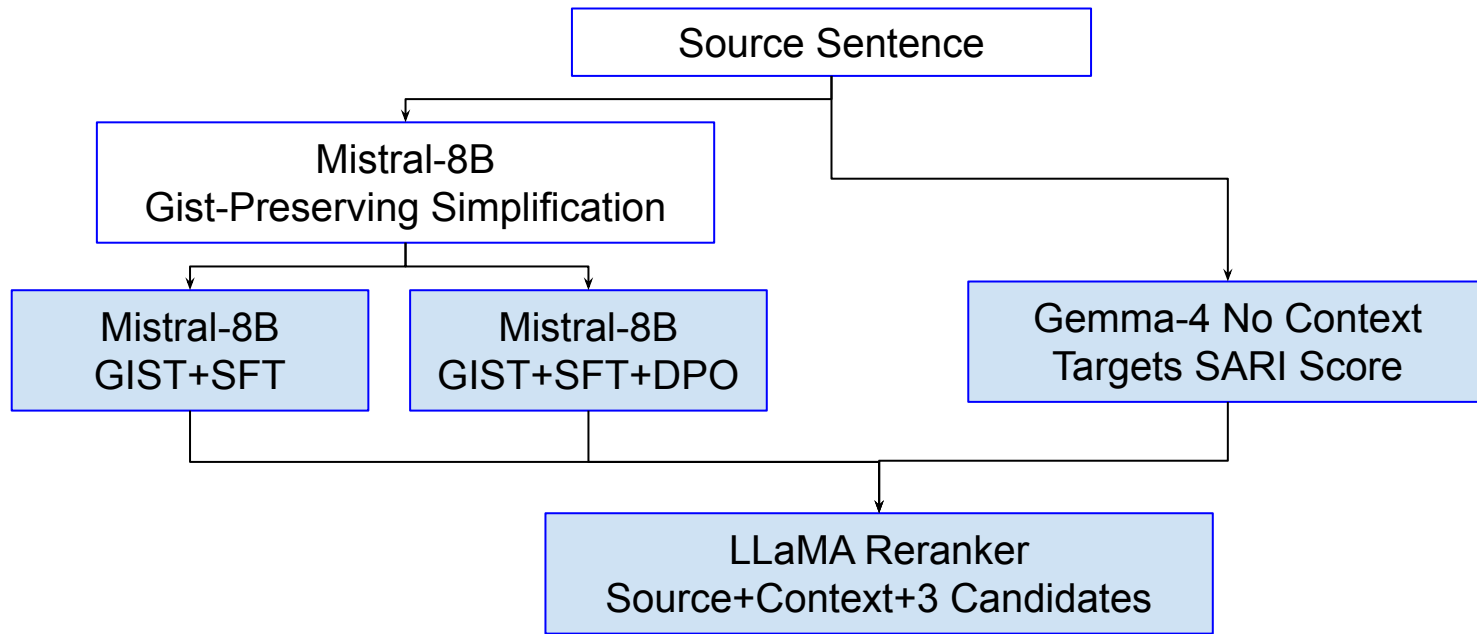
AIIRLab Participation in 2026

We participated in all three tasks:

- Task 1: Simplifying scientific text at the sentence and document level
- **Task 2: Identifying and classifying overgeneration**
- Task 3: Research area classification

Our main approaches explore ensemble and multi-agent pipelines using LLMs and other machine learning models

Task 1.1: Sentence-Level Simplification



- GIST Frame: Agent, Predicate, Outcome, Quantity, Uncertainty. Used for preserving critical information
- Operation Conditioning: Rephrase, split, merge, delete, or copy. Guides the model toward the intended simplification operation.
- Direct Preference Optimization: Uses preferred and rejected simplifications to align the model with meaning-preserving simplification preferences.

Task 1.1: Result

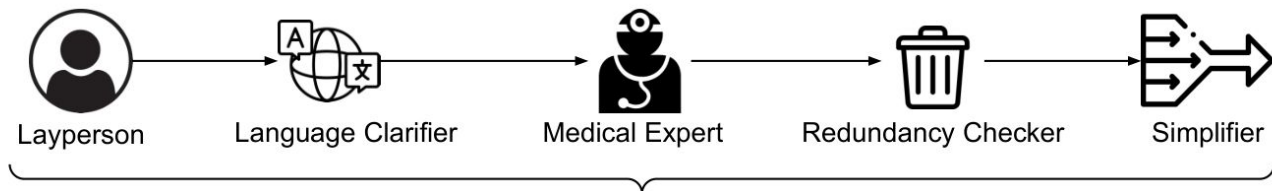
System	SARI	FKGL	BLEU
GSI (Mistral)	42.44	10.45	5.58
DPO (Mistral)	42.68	9.84	5.80
SARI-Based Prompt (Gemma)	41.97	8.66	11.04
Reranker (LLaMA)	46.24	11.36	10.80

- Preference-pair training (DPO) improves over the base (GSI) pipeline
- Combining the three systems with the reranker improves SARI score

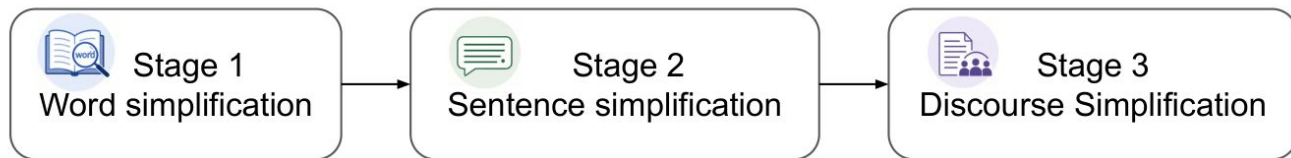
Task 1.2: Document-Level Simplification



Five-Agent
Collaborative Pipeline



Three-Stage
Sequential Pipeline



- Five-Agent Gemma Collaborative Pipeline: Sequential document simplification
- Three-stage LLaMA: Word → Sentence → Discourse simplification
- Atomic Plain-language Instruction Optimization (APIO): Gemma prompting targeting high SARI score
 - Replacing jargon (ADD), preserving GRADE certainty labels and numerical findings exactly (KEEP), and removing verbose structure (DELETE)
- Baseline LLaMA: Zero-shot prompting

Task 1.2: Results

System	SARI	FKGL	BLEU
Five-Agent Pipeline (Gemma)	42.00	13.03	4.04
Three-Stage Pipeline (LLaMA)	41.74	14.31	8.00
Few-Shot Prompt (Gemma)	41.41	13.83	11.83
Zero-Shot Prompt (LLaMA)	42.67	11.06	5.48

Agentic approaches were not successful

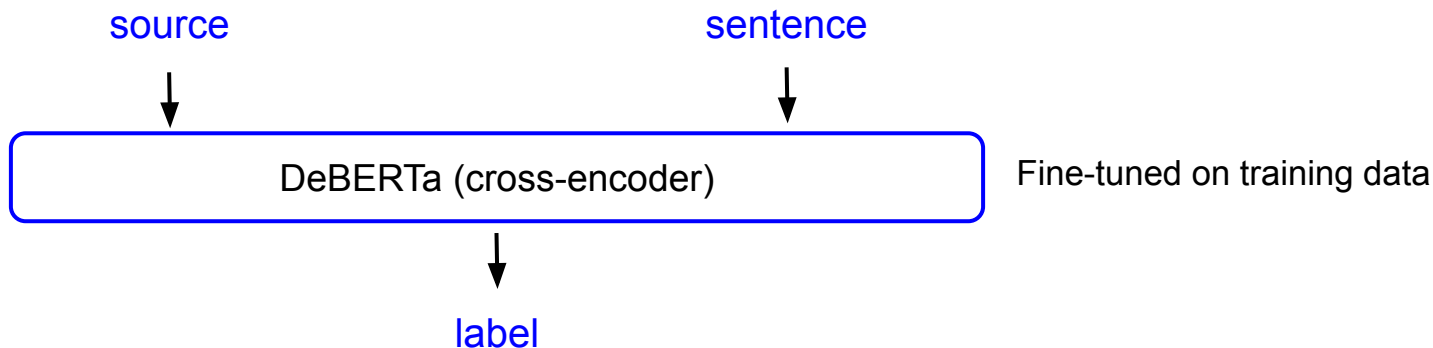
- The Multi-agent Gemma system shows the most aggressive addition and sentence-splitting behavior
- Multi-step LLaMa model produces the highest FKGL

Task 2: Detect and Classify Overgeneration

We identified 24 RCTs involving 3739 participants which were included in the review.

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Training data may contain noise.
Label "None" for this example.



- Three variants of DeBERTa: base, large, enhanced
 - Enhanced variant: binary level (None vs. Error) → fine-grained
- Fine-tuned LLaMA model
- **Ensemble**: Majority voting between 2 DeBERTa and LLaMA



Task 2: Results

System	Sentence F1 (2.1)	Document F1 (2.1)	Accuracy (2.2)
Cross-Encoder (DeBERTa large)	0.8083	0.7481	0.8326
Cross-Encoder (DeBERTa base)	0.7783	0.7243	0.8024
Enhanced Cross-Encoder	0.8486	0.8030	0.8161
Fine-Tuned LLaMA	0.8483	0.7724	0.7693
Ensemble	0.8633	0.8197	0.8269

DeBERTa large outperformed base in every metric

Highest recall with DeBERTa large

- 0.9114 → sentences
- 0.8632 → documents

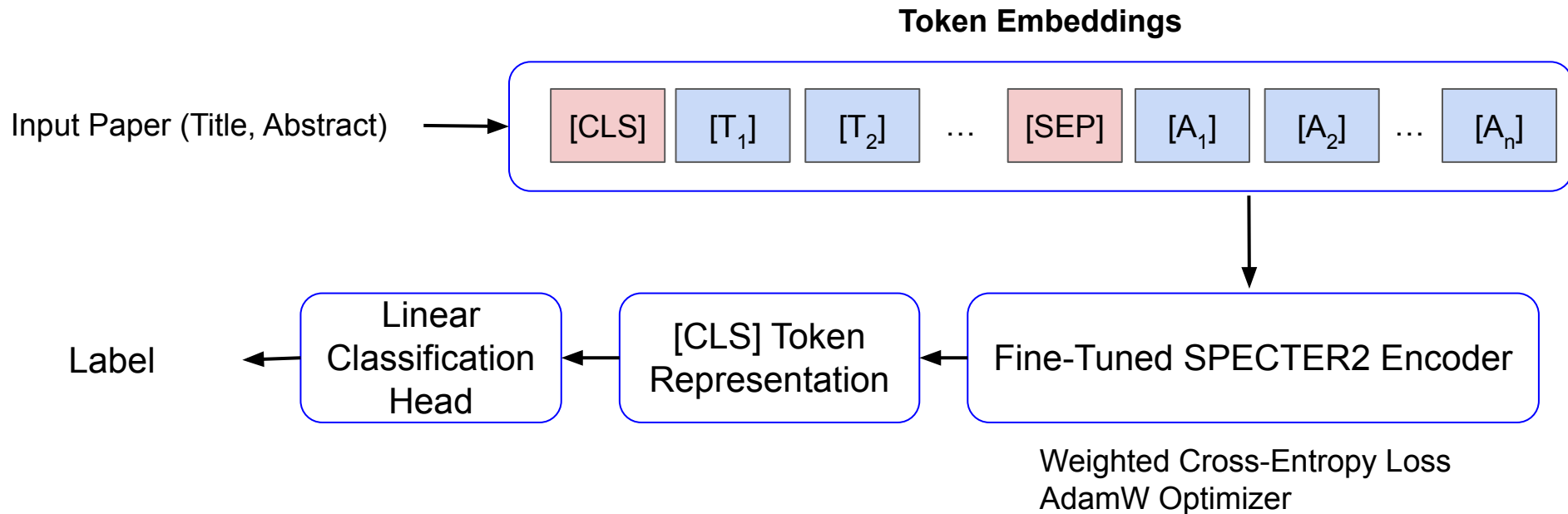
High precision with Fine-tuned LLaMA

- 0.8374 → sentences
- 0.8153 → documents

Ensemble achieved the highest F1

Task 3: Research Area Classification

1. Fine-tuned Scientific BERT Classifier



2. Fine-tuned LLaMA Classifier

Task 3: Results

Exact Match(EM), String Distance (SD), and Embedding Distance (ED)

	Discipline Area			Field Area		
	EM	SD	ED	EM	SD	ED
LLaMA	0.58	0.67	0.77	0.73	0.82	0.87
SPECTER2	0.65	0.74	0.81	0.74	0.83	0.87

Task 3: Results

Exact Match(EM), String Distance (SD), and Embedding Distance (ED)

	Discipline Area			Field Area		
	EM	SD	ED	EM	SD	ED
LLaMA	0.58	0.67	0.77	0.73	0.82	0.87
SPECTER2	0.65	0.74	0.81	0.74	0.83	0.87

Exact Match Results

	Natural Sciences	Life Sciences	Engineering Sciences	Social Sciences	Humanities	Macro Avg.
#Training	9,409	907	1,601	361	23	–
LLaMA	0.92	0.80	0.78	0.82	0	0.66
SPECTER2	0.91	0.82	0.87	0.75	0	0.67

Conclusion & Future Work

We found that ensemble and multi-agent pipelines can outperform single models in both simplification and classification tasks

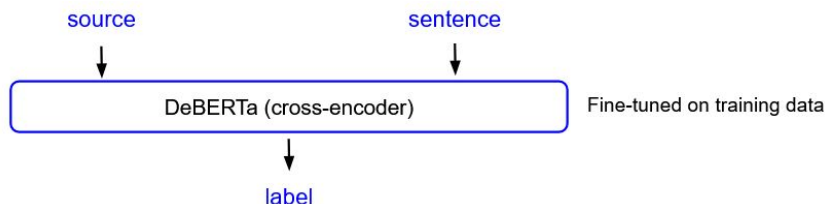
Future work

- Improving factual consistency in document-level simplification
- Using finer-grained reasoning for overgeneration detection
- Exploring classification structures that optimize both field and discipline predictions

Release of gold standards for test sets will help a lot with analysis of our approaches



Many thanks to the SimpleText organizers for all their efforts!



Ensemble DeBERTa + LLaMA for
Detecting and Classifying Overgeneration

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